Investigating the Causal Effects of the ARRA Subsidies on Food Security

A Causal Difference-In Differences Event Study Examining SNAP Benefits on Food Budget Shortfalls

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Cover

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Abstract

This paper estimates the causal effect of the 2013 expiration of the American Recovery and Reinvestment Act (ARRA) SNAP benefit supplements on county-level food security outcomes. Using a Two-Way Fixed Effects event study design spanning 2010–2017 across U.S. counties, we define treatment as above-threshold SNAP utilization rates in the pre-treatment period and estimate dynamic treatment effects on two outcomes: the number of food insecure persons and the Weighted Annual Food Budget Shortfall (WAFBS). We find that treated counties experienced approximately 2,207 additional food insecure individuals in the first post-lapse year, and a persistent increase in WAFBS of up to $1.76 million by year four. A decomposition analysis further reveals that the shortfall increase cannot be fully attributed to the rise in food insecure headcount alone, implying a worsening in the depth of food insecurity beyond its breadth. These findings are directionally consistent with prior literature and suggest that even modest reductions in SNAP benefit levels impose measurable and persistent burdens on high-utilization communities.

1 Introduction

Following the 2008 Financial Crisis, the U.S. Government legislated a massive subsidy package designed to revitalize the economy, named the American Recovery and Reinvestment Act (ARRA). This package introduced in close to $800 billion in total subsidies (1), providing approximately a $20 boost per person in its first year to Supplemental Nutrition Assistance Program (SNAP) benefits (2). Another notable event was the passing of the Healthy, Hunger-Free Kids Act of 2010 (school lunches), which accelerated the consumption of the allocated funding, depleting the resources quicker than expected. Benefits ended at the end of October, 2013 (2). This event is a massive boon for this study as it gets in the way of some anticipation effects, where the program was sunset earlier than the public would have planned for.

The events pertaining to the SNAP benefits boost naturally lends itself to an event study Two-Way Fixed Effects (TWFE) design, where we can identify the causal effect of the small cut in subsidies on food security, alongside other regressands.

2 Literature Review

We survey the literature for similar studies, highlighting key papers pertaining food security and the ARRA lapse. First, Nord & Prell study the onset of the ARRA unlike the expiration of them in this paper, in which they find SNAP eligible households increased their food expenditures by 5.4 pp, alongside decreasing their insecurity by 2.2 pp (3). Nord further expanded upon this with a DiD analysis by studying the decline of SNAP benefits in real terms due to inflation, which finds a decline in food spending and an increase in the SNAP-recipients (4).

Katare and Kim study the effects of the ARRA lapse, as conducted in this paper, albeit with a much more restricted panel of 2012-2014 using a similar DiD methodology (5). Their findings, suggesting a 7.6 pp increase in food insecurity using binary response variables (food insecurity status), is directionally consistent with the findings of this paper.

Beatty and Tuttle study the consumer behavior using the ARRA as an event study, showing that upticks in SNAP benefits increases food expenditure by a larger amount than compared to cash (6). The authors also predict that the lapse of the ARRA will bring upon a downturn in household food expenditure, a finding consistent with that in Section 5.3.

An example from recent literature corroborating the mechanisms argued in this paper is that of Richterman et al., where emergency SNAP increases during COVID are examined to yield similar findings, where SNAP recipients are concluded to be impacted in a harsher manner due to emergency allocation lapses (7).

3 Data

To investigate this project, we create 4 individual panels, merging them into a single dataset. Due to the event study pertaining to the years 2009 (partial) to the treatment date of October 2013, we start the panel (years, in time dimension) in the year of 2010, the first year of full effect. The panel spans all the way to 2017, which provides ample data in the post periods.

For the entity dimension, we examine the data at the county level, which is the most precise level reported common between all of the datasets. Below, each are discussed in detail.

- **American Community Survey Data (ACS)** is used to extract the county level SNAP utilization rates across time. This pertains to the definition of our treatment, which is mathematically defined in the Section 4 (8).
- **Small Area Income and Poverty Estimates (SAIPE)** contains poverty indicators, including the number of food insecure (FI) persons, their rates, alongside these metrics at an age-stratified level (9).
- **Local Area Unemployment Statistics (LAUS)** is compiled by the Bureau of Labor Statistics, and contains data on unemployment rate and labor force, to control for post-2008 economic trends in the event study (10).
- **Map the Meal Gap (MMG)** (11) is maintained by an NGO, *Feeding America* collects data on food insecurity using methodologies described in their report (12). Here, we are primarily concerned with the metric *Weighted Annual Food Budget Shortfall* (WAFBS):
$$WAFBS = FI \times S \times 52 \times \frac{7}{12}$$
where S represents the weekly shortfall of a single person in USD, and $\frac{7}{12}$ is an empirically validated constant that scales the true shortfall (12). The dataset also allows for the access into the "Cost per Meal", which is a metric that derives the true cost to attain a meal using survey results of food secure individuals only.

For our model, we restrict our sample to target the elimination of unbalanced panel contributors, namely counties with ≤ 6 years available, additionally excluding Washington D.C. Below, we also provide a summary statistics table, limited to the mentioned regressands or controls in this paper. Rejected candidate regressands are omitted. The reader can access plots of the same style in this paper in the analysis file if they are interested in the rejected candidate regressands.

Table 1: Descriptive Statistics

	Labor Force	Unemp. Rate (%)	WAFBS (USD)	Num FI Persons
N	6,365	6,365	6,365	6,365
Mean	164,480	6.80	22,963,560	45,599
Std. Dev.	291,404	2.68	43,652,530	85,477
Min	20,139	2.00	2,019,080	4,030
25th Pct.	46,545	4.90	6,384,000	13,230
Median	76,369	6.30	10,351,000	21,020
75th Pct.	166,352	8.30	22,439,000	44,990
Max	5,107,316	29.10	815,943,000	1,749,600

Panel covers county-year observations, 2010–2017. WAFBS = Weighted Annual Food Budget Shortfall.

4 Model

We estimate many regressands with the choice of our two way fixed effects and a pool of viable control variables. All models cohere to the following event study specification:

$$Y_{ct} = \lambda_t + \alpha_c + \sum_{i=-3}^{4} \mathbb{1}\{i \neq -1\}\beta_i D_{ct} + \epsilon_{ct}[1]$$

As required by the version of the Neyman-Rubin Potential Outcomes Framework introduced in this class, we dichotomize the continuous variable of SNAP utilization rates as the treatment, with higher utilization counties being defined as "treated". We characterize this as the following:
Let $\hat{F}_c$ be the empirical cumulative mass function of all SNAP utilization rates across counties. Note that because there are 3 pre-treatment years, we aggregate over all 3 years using a simple mean to generate one point per county. Define $\theta \in [0, 1]$ s.t. all counties c if c is above the θ quantile of the distribution $\hat{F}_c$, then the county is considered treated:

$$D_{ct} = \mathbb{1}[c \text{ above } \theta \text{ quantile of } \hat{F}_c]$$

noting that the treatment is time invariant. We note this as a potential threat to validity, where we could be defining the treatment to be correlated with economically worse-off counties to begin with, which would be problematic had these counties been on a different trend. For this reason, we are diligent in providing falsification tests.

[1] Errors are clustered at the county level.

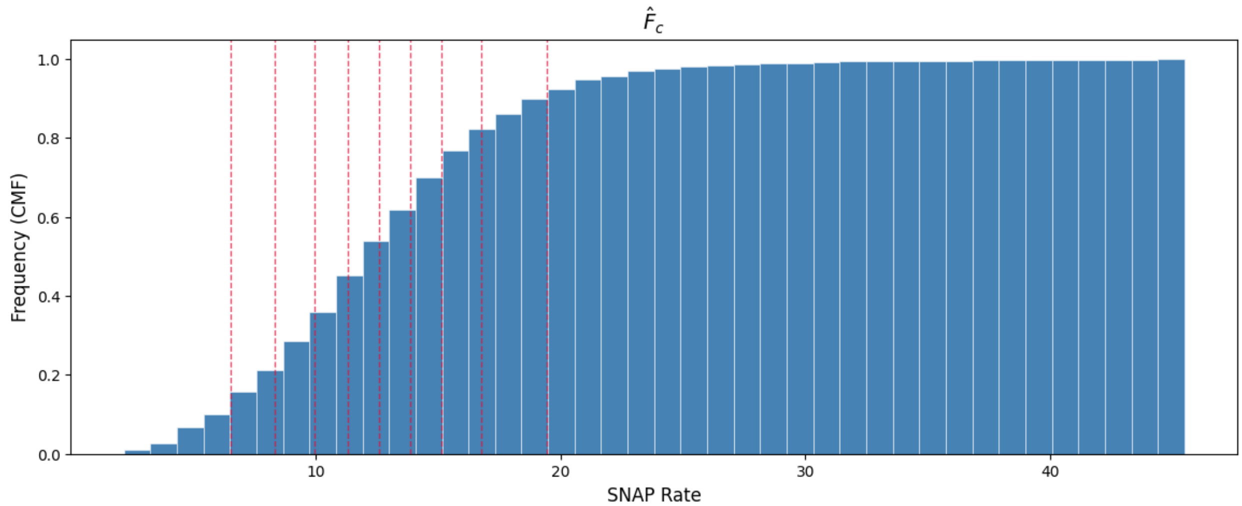

Figure 1: Empirical Distribution $\hat{F}_c$ with each 10th Quantile Marked

The selection of the parameter θ is semi-arbitrary, where we argue that $\theta = 0.3$ is a "good enough" threshold. We later show that the results are statistically robust across different $\theta \in [0.2, 0.45]$. In selecting θ, it also makes sense to not define the parameter as too small or too large, where the balance of the control and treatment group sizes is a direct consequence of it.

We further argue that because the ARRA expiration is a common event, there is no case for staggered adoption. This rules out the existence of the "forbidden comparison"s in the Goodman-Bacon decomposition.

5 Results

We consider two viable regressands that conform to the parallel trends assumption of the DiD framework.

5.1 Number of FI Persons

The first candidate is the number of FI persons, controlling for the unemployment rate, provided in Figure 2.

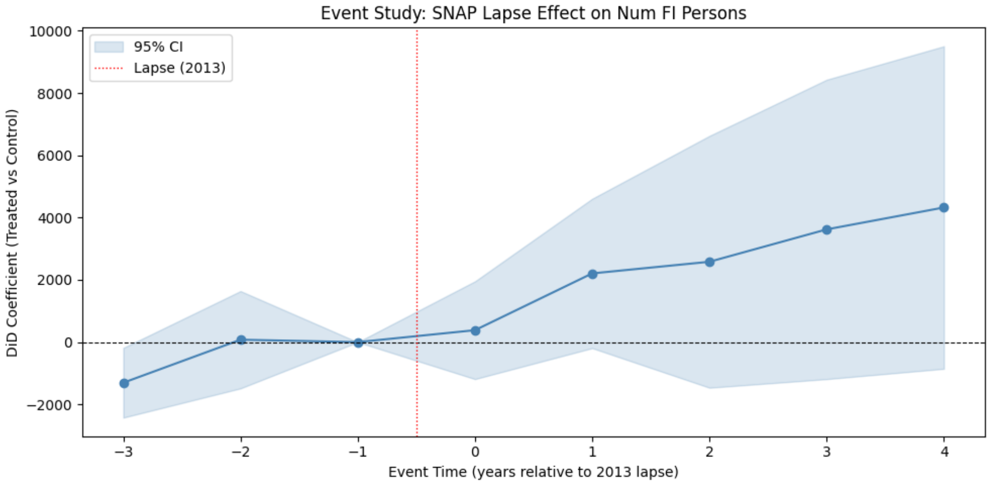

Figure 2: Number of FI Persons Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Below is the table of regression results. First, we make note of the fact that the 0th time period has no meaningful effect. As a recurring theme in the following sections, we make note of the fact that the treatment only applies at the final month of the 0th year, making any regression with significant year 0 coefficient highly dubious. We can notice an immediate effect in year 1 (only marginally statistically significant, $p = 0.07$), where everything else held constant, the first time step sees an expected 2207 additional food insecure persons at a county level where treated. The remaining time steps are statistically insignificant. We also highlight the significant control of the unemployment rate, where within the same county at the same time, a 1% increase in unemployment is associated with an expected increase of 2672 food insecure individuals.

Table 2: Event Study Results for Number of FI Persons Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Variable	**Coef.**	**Std. Err.**	**t**	**P> |t|**	**[0.025**	**0.975]**
`did_evt_-3`	-1305.0627^{**}	571.2038	-2.28	0.0224	-2424.8456	-185.2798
`did_evt_-2`	77.3812	797.0296	0.10	0.9227	-1485.1086	1639.8709
`did_evt_0`	383.1136	800.8639	0.48	0.6324	-1186.8931	1953.1202
`did_evt_1`	2206.5367^{*}	1224.0087	1.80	0.0715	-192.9992	4606.0725
`did_evt_2`	2580.2147	2063.1419	1.25	0.2111	-1464.3504	6624.7799
`did_evt_3`	3620.7681	2452.5657	1.48	0.1399	-1187.2202	8428.7565
`did_evt_4`	4323.7989	2642.0558	1.64	0.1018	-855.6641	9503.2619
Unemployment Rate (%)	2671.7289^{***}	695.5873	3.84	0.0001	1308.1057	4035.3521

Lastly, we acknowledge the violation of the PT assumption in year -3 based on the full coefficient table. We run a PT violation test by restricting the data to the pre-treatment periods only yields results which are not sufficient to reject PT, though the power of this secondary test is suspect due to the low number of pretrend years. Because we rely on the directionality on these findings to guide Section 5.3, we note this as a limitation of this paper and therefore opt to interpret this result as a directional signal, leaning more heavily on the conclusions of the later sections.

Table 3: Pre-Treatment Falsification Test Results for Number of FI Persons Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Variable	Coef.	Std. Err.	t	P> |t|	[Lower CI	Upper CI]
`did_evt_-3`	−627.43	568.72	−1.1032	0.2701	−1742.9	488.09
`did_evt_-2`	523.74	810.84	0.6459	0.5184	−1066.7	2114.2

5.2 Weighted Annual Food Budget Shortfall (WAFBS)

A similar result can be observed in the WAFBS metric as the regressand with some differences. First, the PT violation based on the full set of regressands is no longer present, a property maintained in the falsification tests below. Notably, we uncover some dynamic effects that are significant at the 4th year as well, with the weighted annual food budget shortfall increasing by 843,000 and 1,760,000 USD at years 1 and 4 where treated, respectively, in a statistically significant manner at $\alpha = 0.05$. In a similar fashion, unemployment is also a highly significant control, with a 1% increase in unemployment in the same county and year, associated with a 745,000 USD increase in the shortfall metric.

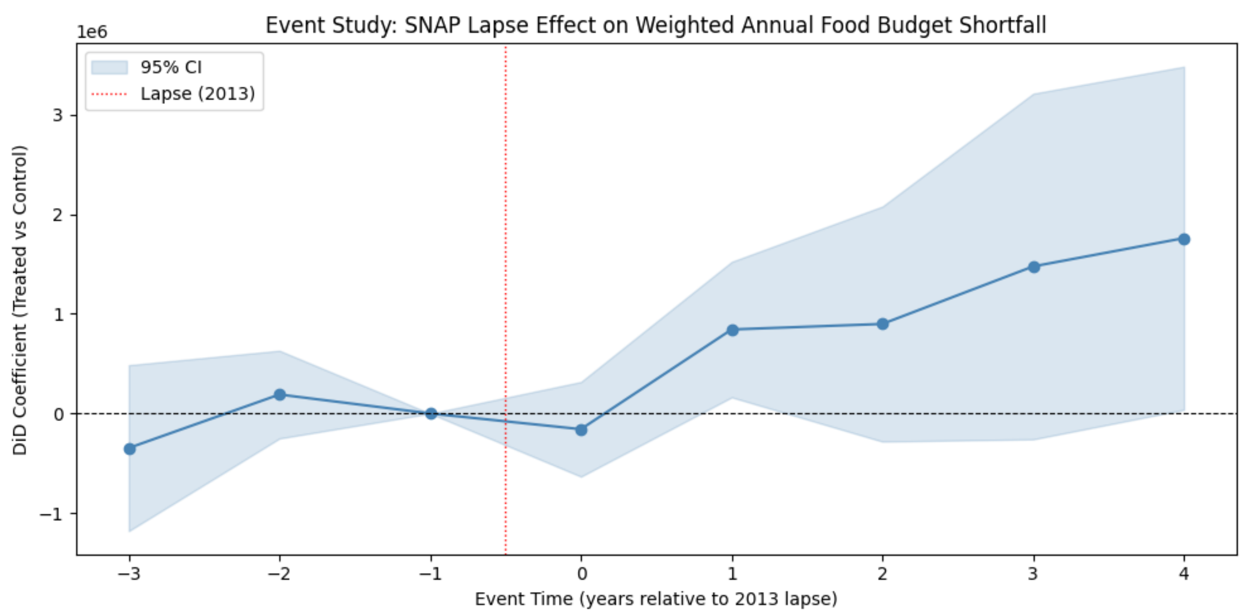

Figure 3: WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Table 4: Event Study Results for WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Variable	Coef.	Std. Err.	t	P> |t|	[0.025	0.975]
did_evt_-3	-3.47×10^5	4.25×10^5	-0.82	0.4136	-1.18×10^6	4.85×10^5
did_evt_-2	1.90×10^5	2.25×10^5	0.85	0.3972	-2.50×10^5	6.30×10^5
did_evt_0	-1.58×10^5	2.42×10^5	-0.65	0.5147	-6.33×10^5	3.17×10^5
did_evt_1	$8.43 \times 10^{5**}$	3.47×10^5	2.43	0.0151	1.63×10^5	1.52×10^6
did_evt_2	8.98×10^5	6.02×10^5	1.49	0.1356	-2.82×10^5	2.08×10^6
did_evt_3	$1.48 \times 10^{6*}$	8.85×10^5	1.67	0.0954	-2.59×10^5	3.21×10^6
did_evt_4	$1.76 \times 10^{6**}$	8.78×10^5	2.00	0.0452	3.78×10^4	3.48×10^6
Unemp. Rate (%)	$7.45 \times 10^{5***}$	1.95×10^5	3.82	0.0001	3.62×10^5	1.13×10^6

Table 5: Pre-Treatment Falsification Test Results for WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Unemployment Rate

Variable	Coef.	Std. Err.	t	P> |t|	[Lower CI	Upper CI]
did_evt_-3	-1.52×10^5	5.09×10^5	-0.2989	0.7651	-1.15×10^6	8.46×10^5
did_evt_-2	3.15×10^5	2.50×10^5	1.2608	0.2076	-1.75×10^5	8.04×10^5

5.3 WAFBS Increase Decomposition

We then ask the following question, based on the WAFBS metric: Considering the fact that there is evidence of an increase in both the FI individual count, and the WAFBS, is the increase of WAFBS solely due to the FI increase?

$$WAFBS = FI \times S \times 52 \times \frac{7}{12}$$

In order to answer this question, we propose a third specification, where we regress WAFBS again, but controlling for the mediator FI. We note that because the number of FI persons is a post-treatment outcome, controlling for it introduces a mediator issue. Therefore, we interpret this specific specification as a descriptive decomposition of the shortfall's depth versus breadth, rather than a strict causal outcome. We also substitute unemployment rate for labor force, an empirically validated decision.

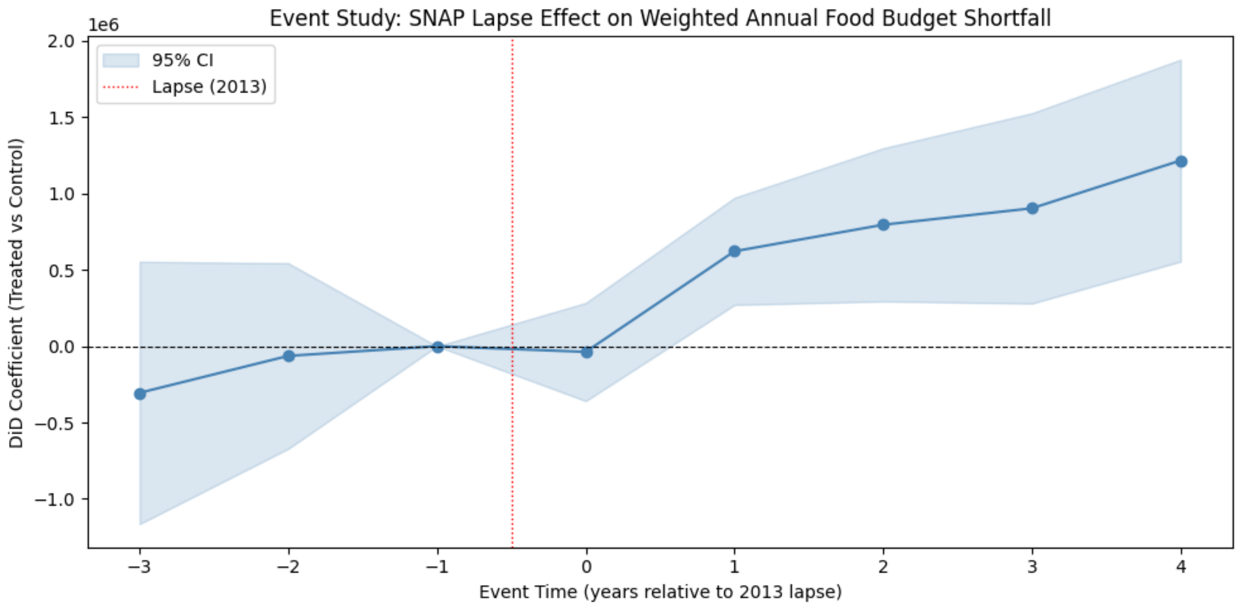

Figure 4: WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Number of FI Persons and Labor Force

When we examine the coefficients table, we can see an increasing dynamic effects trend, where relative to year -1, the shortfall increases relative to the controlled group, by 621,000, 795,000, 903000, 1,220,000 USD from years 1 to 4 in expectation. This signals that the effects are persisting and laggy to manifest fully. Further, we note the statistical strength of the regression, with every "desirable" regressand significant even at an $\alpha = 0.01$ level.

Table 6: Event Study Results for WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Number of FI Persons and Labor Force

Variable	**Coef.**	**Std. Err.**	**t**	**P>|t|**	**[0.025**	**0.975]**
`did_evt_-3`	-3.05×10^5	4.38×10^5	-0.70	0.4857	-1.16×10^6	5.53×10^5
`did_evt_-2`	-6.35×10^4	3.10×10^5	-0.21	0.8375	-6.71×10^5	5.44×10^5
`did_evt_0`	-3.73×10^4	1.64×10^5	-0.23	0.8201	-3.59×10^5	2.84×10^5
`did_evt_1`	$6.21 \times 10^{5***}$	1.79×10^5	3.47	0.0005	2.71×10^5	9.72×10^5
`did_evt_2`	$7.95 \times 10^{5***}$	2.56×10^5	3.11	0.0019	2.94×10^5	1.30×10^6
`did_evt_3`	$9.03 \times 10^{5**}$	3.18×10^5	2.84	0.0045	2.80×10^5	1.53×10^6
`did_evt_4`	$1.22 \times 10^{6***}$	3.37×10^5	3.61	0.0003	5.55×10^5	1.88×10^6
Num FI Persons	$3.10 \times 10^{2***}$	2.15×10^1	14.43	0.0000	2.68×10^2	3.52×10^2
Labor Force	$1.97 \times 10^{2***}$	2.80×10^1	7.02	0.0000	1.42×10^2	2.52×10^2

The pre-trend falsification tests also pass as seen below:

Table 7: Pre-Treatment Falsification Test Results for WAFBS Relative to Control Group, $\theta = 0.3$, Controlling for Labor Force and Number of FI Persons

Variable	**Coef.**	**Std. Err.**	**t**	**P>** $|$**t**$|$	**[Lower CI**	**Upper CI]**
`did_evt_-3`	-7.09×10^5	5.67×10^5	-1.2514	0.2110	-1.82×10^6	4.03×10^5
`did_evt_-2`	-2.74×10^5	3.78×10^5	-0.7240	0.4691	-1.02×10^6	4.68×10^5
Labor Force	412.65^{***}	93.490	4.4139	0.0000	229.27	596.03
Num FI Persons	258.93^{***}	53.384	4.8503	0.0000	154.22	363.64

We also call attention to the coefficient on the "Num FI Persons" at 310 USD. This signals that increasing the number of FI persons by one brings about an increase of 310 USD in the shortfall increase. Recalling the formula,

$$\underbrace{WAFBS}_{310} = \underbrace{FI}_{=1} \times S \times 52 \times \frac{7}{12}$$

this change must originate from $S \times 52 \times \frac{7}{12}$. Solving for S, this yields $S = 10.21$ USD, where S was quoted as a weekly shortfall. Adjusting time frames, this yields the shortfall as 1.46 USD/day, or 43.77 USD/mo. Of course, this S is indeed a variance weighted average of county-specific S values, so we call for caution against the over-interpretation of this result. To contextualize these results, a Congressional Research Service Report puts the estimated losses in SNAP benefits due to the ARRA lapse as the below table, where household size 3 is the most common (13). Note that the prior result is that of S, which tracks dollar losses per person at the county level, whereas the table is specific to the subpopulation of the SNAP recipients - which makes any direct comparisons between the 43.77 USD figure and the table invalid.

Table 8: Estimated SNAP Subsidy Losses on a Per-Person Basis

Household Size	**Monthly Dollar Loss Per Person**
1	24
2	22
3	21
4	20

6 Robustness

We begin by investigating the effect of θ on the validity of our regressions. By varying θ from 0.1 to 0.65 we observe that our result would hold for the approximate range of $\theta \in [0.2, 0.45]$. A too high or too low of θs have problems associated regardless of these findings. If θ is too high, then the control group begins to include the SNAP dependent counties. Based on the $\hat{F}_c$ distribution, when the 60th decile is reached, SNAP utilization is already in the post-inflection point of the empirical CMF, contaminating the controls with units that are almost certainly treated by any definition. Similarly, if θ is too low, then the treatment group begins to be contaminated with the untreated - the effects of both of which can readily be observed with the widening confidence intervals as we drift away from $\theta = 0.3$.

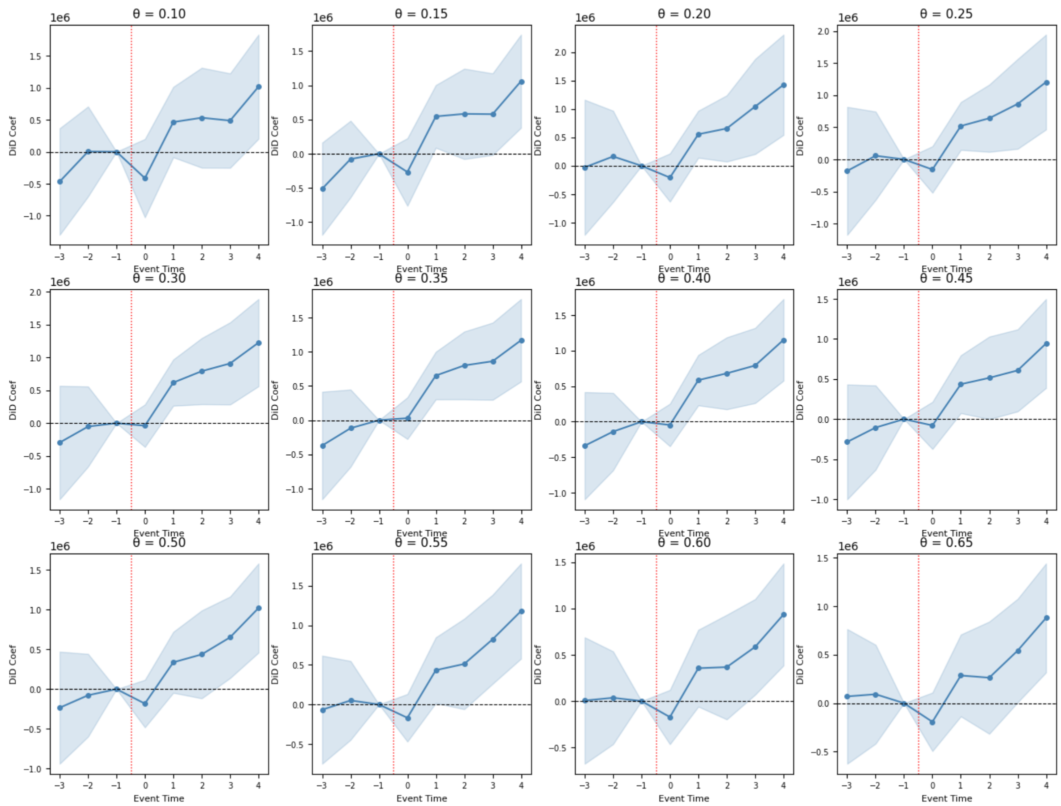

Figure 5: Effects of Varying θ on WAFBS Relative to Control Group, Controlling for Number of FI Persons and Labor Force

Next, due to the limitation of the pre-treatment periods, we propose that running time-based placebos lacks statistical merit, as with 3 periods, no result is capturing true trends even if they exist. Instead, we propose the permutation placebo: for 500 iterations, we assign the treatment randomly instead of the θ methodology, and record the average post-treatment effects. These trials generate a null-distribution, which our result should sit well outside of. Indeed, the results show that the initial findings sit at the tail of the distribution, yielding strong evidence that the effect occurs far from random chance.

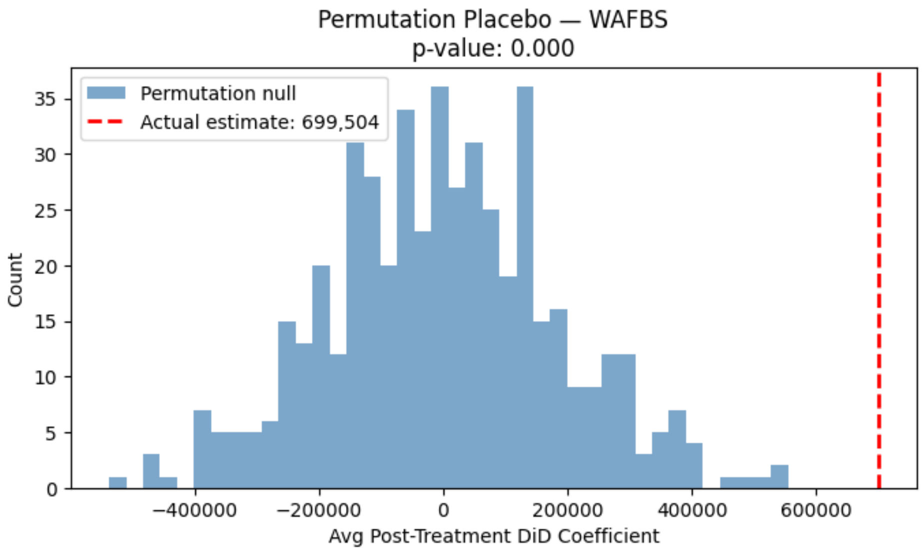

Figure 6: Real Outcome Against Plotted Empirical Null-Placebo Distribution

7 Conclusion

In this paper, we set out to investigate the causal effects of the 2013 SNAP ARRA benefits lapse on a county level. We found evidence of an increase in both the number of food insecure individuals, and the food insecurity of the average food insecure individual itself, with dynamic and persistent effects spanning to 2017 and possibly beyond (limitation of data availability). These findings go to validate the predictions of Beatty & Tuttle, while also conforming to the directionality of Katare & Kim.

As acknowledged in the results and robustness sections, some limitations persist, even if mitigated by other methods. First, while the PT violation in Section 5.1 is an ongoing concern, we point to the later sections, especially that of Section 5.3 which is the main finding of this paper, where PT violations are no longer present. Similarly, while the availability of pre-trend year data is limited, thus hampering the possibility of pre-trend violation tests' power and ruling out the feasibility of placebos, we use the simulated random-treatment assignment to assure our readers about the statistical improbability of our findings occurring by random chance.

On the topic of policy based recommendations, we draw attention to the modesty of the ARRA lapse generating changes that aren't absorbed by households. Instead, persistent deterioration of food security that, within the confines of interpretability outlined in Section 5.3, appears to rival, or perhaps exceed, the lapse itself highlights the impact of such small contributions to lower-income households.

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